

National University of Computer and Emerging Sciences, Lahore Campus



Course Name: Engineering Management
 Program: BS(Electrical Engineering)
 Duration: 150 Minutes
 Paper Date: 20-December-17
 Section: ALL
 Exam Type: Final

Course Code: MG330
 Semester: Fall 2017
 Total Marks: 50
 Weight: 50%
 Page(s): 2

Student Name: [Redacted]
 Instruction/Notes: [Redacted]

Roll No. [Redacted] Section: [Redacted]

from new page, show all steps, and assume if something is missing in paper.

Q #1: Draw a PERT diagram and find the critical path

(10 marks)

Activity	Activity Description	Immediate Predecessors	Estimated Duration
A	Excavate	-	2 Weeks
B	Lay the foundation	A	4 Weeks
C	Put up the rough wall	B	10 Weeks
D	Put up the roof	C	6 Weeks
E	Install the exterior plumbing	C	4 Weeks
F	Install the interior plumbing	E	5 Weeks
G	Put up the exterior siding	D	7 Weeks
H	Put up the exterior painting	E, G	9 Weeks
I	Do the electrical work	C	7 Weeks
J	Put up the wall board	F, I	8 Weeks
K	Install the flooring	J	4 Weeks
L	Do the interior painting	J	5 Weeks
M	Install the exterior fixtures	H	2 Weeks
N	Install the interior fixtures	K, L	6 Weeks

Q #2: Define the objective of forecasting, what are the steps by which we can implement forecasting.

(5 marks)

Q #3: Describe the five possibilities which range from no structure to a totally separate project organization.

(10 marks)

Q #4: According to Pinto what are the most important influences promoting a project management?

(5 marks)

Q #5: Name the five life-cycle phases to classify Costs.

(5 marks)

Q #6: Describe the steps to be carried out to develop LCC model once the agreement is reached on the types of analyses. CBS

(5 marks)

Q#7: One of four new techniques, or the current method, can be used to control mildly irritating chemical fume leakage into the surrounding air from a mixing machine. The estimated costs and benefits (in the form of reduced employee health costs) are given below for each method. Assuming that all methods have a 10-year life with zero salvage value, determine which one should be selected, using a MARR of 15% per year and the incremental B/C method. (10 marks)

	Technique			
	1	2	3	4
Installed cost, \$	15,000	19,000	25,000	33,000
AOC, \$/year	10,000	12,000	9,000	11,000
Benefits, \$/year	15,000	20,000	19,000	22,000